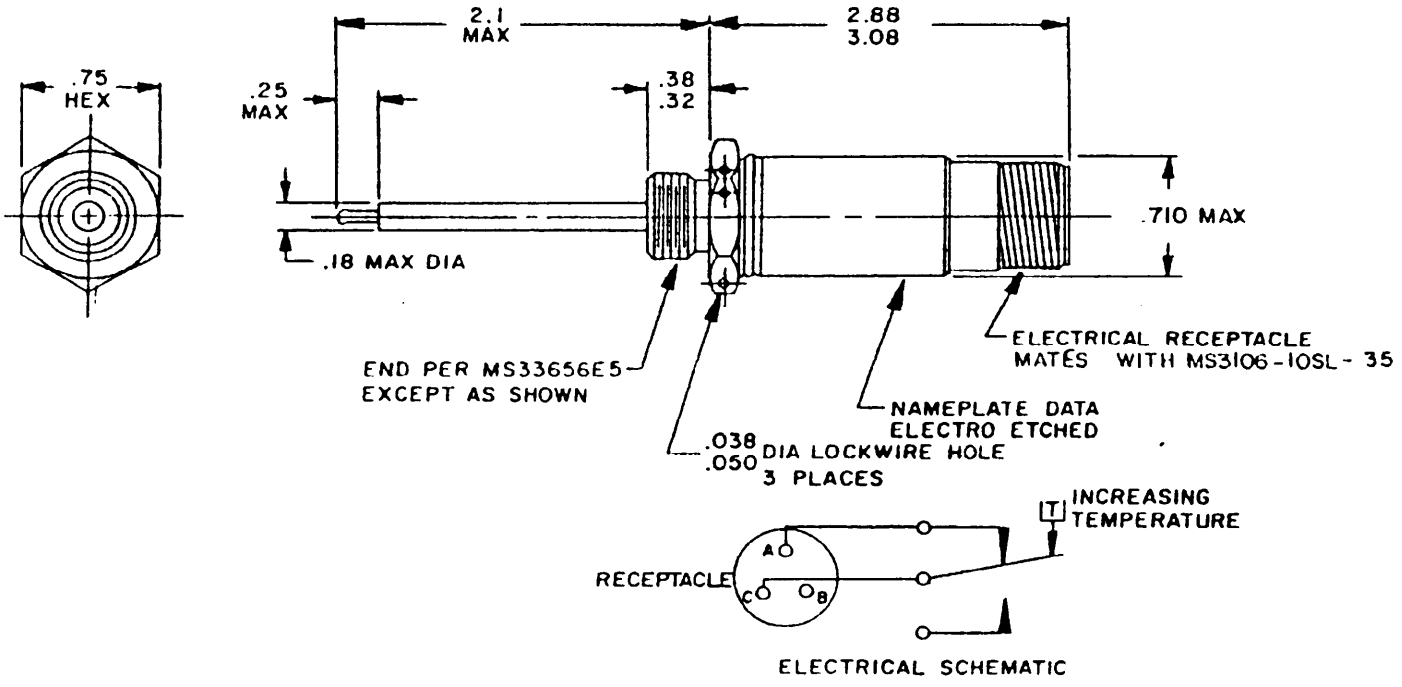
MILITARY SPECIFICATION SHEET

SWITCH, THERMOSTATIC (VOLATILE LIQUID), HERMETICALLY SEALED, SINGLE POLE DOUBLE THROW (SPDT), FAST RESPONSE TIME, LOW LEVEL TO 1.5 AMPERES

This specification is approved for use within the Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the switch described herein shall consist of this specification and the latest issue of MIL-S-28827.

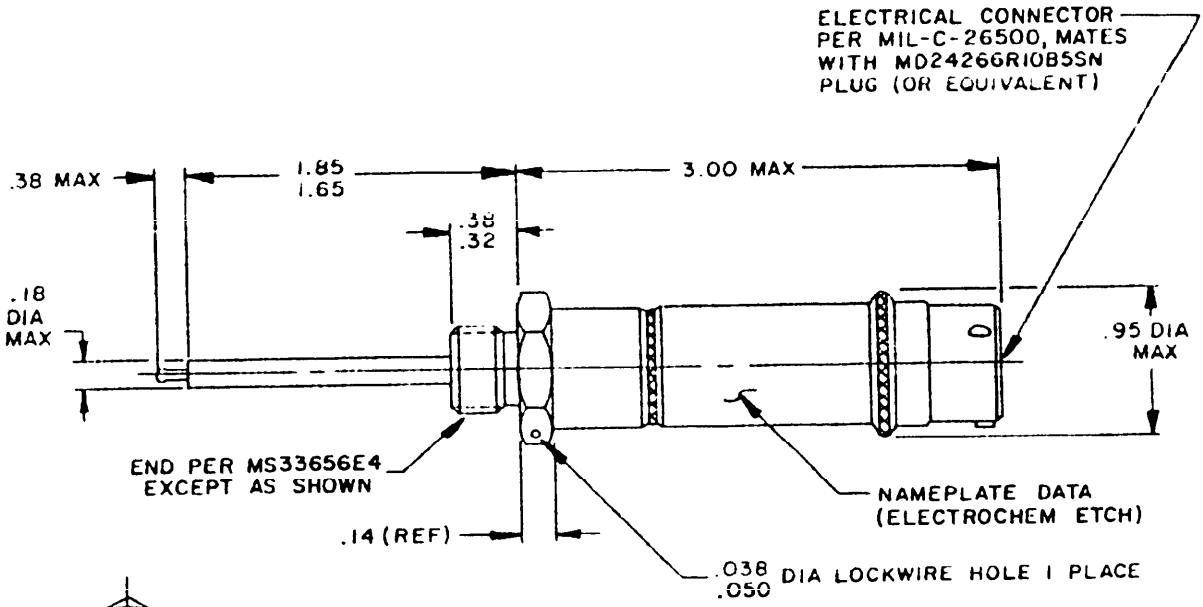


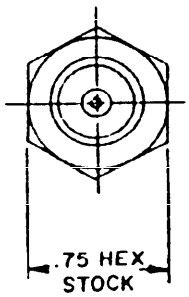
Configuration X.

FIGURE 1. Dimensions and configurations.

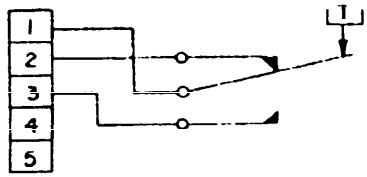
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Inches	mm
.038	0.97
.050	1.27
.14	3.6
.18	4.6
.25	6.4
.32	8.1
.38	9.7
.710	18.03
.75	19.1
.95	24.1
1.65	41.9
1.85	47.0
2.1	53
2.88	73.2
3.00	76.2
3.08	78.2



ELECTRICAL SCHEMATIC

Configuration 4.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm).

FIGURE 1. Dimensions and configurations - Continued.

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REQUIREMENTS

Dimensions, configurations, and electrical schematic: See figure 1.

Weight: 3 ounces.

Operating point tolerances: See tables III, IV, or V as applicable.

Proof pressure: 500 psig.

Proof temperature: 100°F above the maximum increasing set point.

Time constant: Not to exceed 10.0 seconds in 75 centistokes oil at 150 feet/minute.

Electrical ratings:

28 V dc open circuit voltage at 1.5 amperes inductive.

30 millivolts maximum open circuit voltage at 10 milliamperes.

Seal: Electrical chamber hermetic.

Media: Air or any media compatible with wetted materials.

High temperature (ambient operating and nonoperating): 360°F maximum.

Low temperature (ambient operating and nonoperating): -65°F.

Altitude: 80,000 feet.

Endurance, 25,00 cycles: For low level endurance, contact resistance shall not exceed 100 milliohms either during or after the test.

Low level: Applicable when specified by part number.

Contact load: Each switch contact shall make, carry, and break a resistive load of 10 milliamperes maximum at an open circuit voltage of 30 millivolts maximum dc or peak ac. Both normally open and normally closed contacts shall be loaded. Contacts shall be connected to individual loads.

When tested there shall be no failures. A failure shall be an individual circuit contact resistance exceeding 3.0 ohms on any single contact closure.

Qualification:

Two additional samples are required for low level circuit endurance test. Note: Do not exceed 6 V dc or peak ac, 10 milliamperes during inspection of product for delivery when testing low level switches.

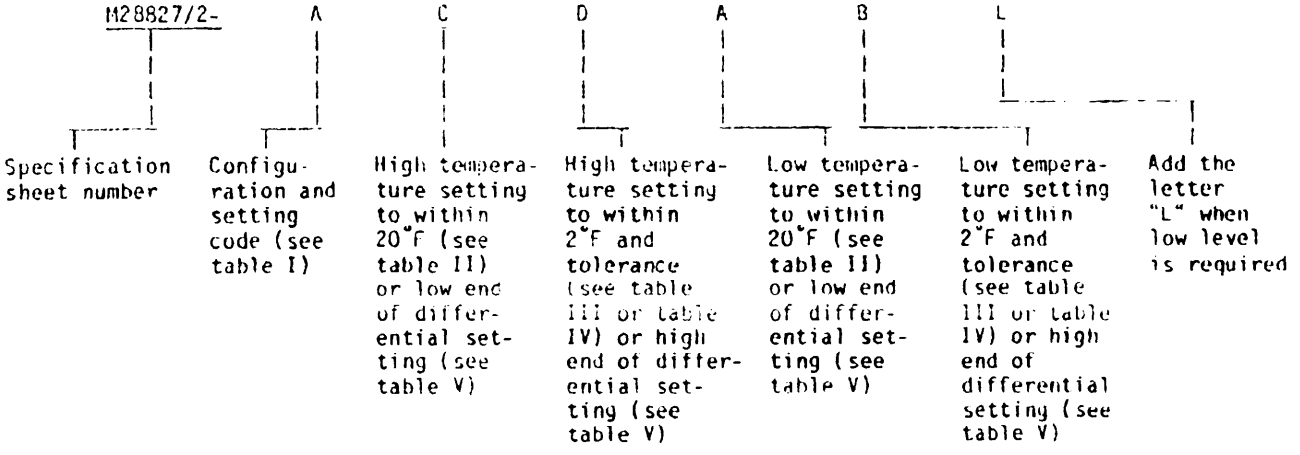
Materials: All wetted materials shall be 300 series stainless steel.

Part number information:

Note: Design limitations (actuation values and tolerances, deadband and deactuation values and tolerances) should be coordinated with a manufacturer(s) listed on the QPL for this sheet before specifying a particular "M" number. The fact that operating characteristics can be coded does not necessarily mean that it can be manufactured or acquired.

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Part number shall be assigned as illustrated below:



Example of part numbers:

M28827/2-ACDAB identifies a switch of configuration X which actuates on increasing temperatures at 146°F ±5°F and deactuates at 102°F ±5°F

M28827/2-EBCDE identifies a switch of configuration X which actuates by 124°F maximum and deactuates by 168°F maximum.

M28827/2-JDBEH identifies a switch of configuration Y which actuates at 162°F ±5°F with a 20°F/35°F differential temperature.

M28827/2-JDBEHL identifies a switch of configuration Y with low level circuit capabilities which actuates at 160°F ±50°F with a 20°F/35°F differential temperature.

TABLE I. Code characters for combinations of configurations and temperature setting modes.

High setting	Low setting	Configuration	
		X	Y
At 1/	At 1/	A	G
At 1/	By 2/	B	H
At 1/	Differential 3/	C	J
By 2/	At 1/	D	K
By 2/	By 2/	E	L
Differential 3/	At 1/	F	M

1/ Setting values are designated by characters from tables II and III.

2/ Setting values are designated by characters from tables II and IV.

3/ Setting values are designated by characters from table V.

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TABLE II. Temperature setting to within 20°F.

Temperature	Code	Temperature	Code
100	A	260	J
120	B	280	K
140	C	300	L
160	D	320	M
180	E	340	N
200	F	360	P
220	G		
240	H		

TABLE III. Temperature setting within 2°F and tolerance.

Unit	Tolerance		
	±5	±10	±25
0	A	L	W
2	B	M	X
4	C	N	Y
6	D	P	Z
8	E	Q	1
10	F	R	2
12	G	S	3
14	H	T	4
16	J	U	5
18	K	V	6

TABLE IV. Temperature setting to within 2°F.

Unit	Maximum or minimum
0	A
2	B
4	C
6	D
8	E
10	F
12	G
14	H
16	J
18	K

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TABLE V. Code letters for differential temperature settings.

Code letter	Differential temperature (°F)
A	0
B	5
C	10
D	15
E	20
F	25
G	30
H	35
J	40
K	45
L	50
M	60
N	75

Preparing activity:
Navy - EC

Agent:
DLA - FS

(Project 5930-N583-1)

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